A Selected proof routes

Target, law, loss, feasible family
Original units and M

Weak-error route
Equality counts as wrong
$R^s = [A^s + \tau(M - N)]/M$

Own-risk optimum
Eligible competitors
Same law and loss
Same family

Count hull
(N, A)
Fixed family
Whole reports

Extra for half cost:
complementarity; full $\mathcal{A}_n$
all strict children feasible

Equality-neutral route
Strict error + distinct withholding
$R^s_{\mathrm{TA}} = [A^s_{\mathrm{TA}} + \tau(T - Q)]/M$

Guarded minimizer
exists
Full at-most G
No extras
Retain pairs
Existence only

Distinct-mass
hull (Q, A)
Fixed family
Whole reports

Extra for half cost:
full-profile class domain
all class-strict children feasible
not guard existence alone

Selected routes, not a complete proof graph.
No criterion independence is assumed.

B Why every required refinement is optimal

One law, one loss and one family
g^* is an EXACT minimizer at $\tau = 1/2$. EVERY required child is feasible.

WEAK: full unit refinements
$\bar{e}_{ij} + \bar{e}_{ji} = 1$
Full unrestricted $\mathcal{A}_n$
All strict unit-order children
$q = M - N(g^*)$

NEUTRAL: full class refinements
$c_{ij} + c_{ji} = \delta_{ij}$
Full active-profile equality classes
All class-strict children feasible
Original unit-pair multiplicities
$q = T - Q(g^*)$

Uniform average over
ALL required children
Mean added error $= q/2$
Withholding reduction $= q/2$
Mean $R(r) = R(g^*) = R^*$

Exact optimum + child feasibility
$R(r) - R^* \geq 0$ for every child

Uniform mean of all child gaps
$\frac{1}{L}\sum_{\ell=1}^{L}[R(r_\ell) - R^*] = 0$

Every required child: $R(r) = R^*$

Endpoints: weak $N = M$; neutral $Q = T$.
Neutral raw N need not equal M.
Guarded $T = 0$: one class, all-withhold;
point frontier; Q/T undefined.

Uniform averaging is a proof device.
Caps, fixed orders or protections need
separate arguments. T11 is not this
endpoint. No arbitrary refinement
guarantee for an approximate optimizer.